

Likhit Vyas Yarramsetti

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EXPERIENCE

Software Engineer @ Visa

Denver, CO | Nov 2025 - Present

- Took ownership of a major modernization effort of backend services for Card & Account Maintenance tools; migrating legacy SOAP APIs to RESTful APIs using **C#**, **ASP.NET Core Web API**, and **SQL Server**, with production observability in **Splunk** and **OpenTelemetry**
- Operated distributed OLTP databases in **SQL Server**, supporting high volume transaction systems (16 million+ queries per day)

Software Engineer @ Quantiperm

Chandler, AZ | Jun 2025 - Nov 2025

- Designed, implemented, and optimized concurrent multi-threaded **C++** embedded systems code and **RESTful APIs** on Raspberry Pis for real-time metabolic monitoring and inline carbonation systems, all in **Linux**-based environments for development and deployment
- Engineered system communication layers & control loop systems with **GPIO**, **PWM**, **I²C**, implemented **Modbus** interfaces over **TCP**
- Developed touchscreen and web UIs with **React.js** to monitor these systems, enabling real-time data visualization
- Optimized **React** UIs using tools like **react-scan**, re-engineering **state management** and eliminating unnecessary re-renders, achieving near-instant input responsiveness and significantly improving GUI performance, ensuring low-latency monitoring
- Leveraged AI-assisted **IDE** tools like **Cursor** and **Copilot** to speed development, without compromising code quality

Software Engineer Intern @ Edwards Lifesciences

Irvine, CA | May 2024 - Aug 2024

- Collaborated in an **Agile Scrum** environment utilizing the **C# .NET ecosystem** and **Azure DevOps** to develop the software inside **HemoSphere**: a hemodynamic patient monitoring system used in 5000+ hospitals around the world
- Employed **C#**, **WPF**, and **gRPC**, with **MVVM** approach to implement the GUI and algorithm integration of a new feature: the detection of **SAS (Severe Aortic Stenosis)** in patients
- Sole developer of mock SAS server: a test tool developed with **gRPC** and **Python** used by QA to test the SAS Detection feature
- Crafted a **Python** automation script that saved 70+ hours of manual labor by employing **natural language processing** to add trace relations for over 5000 SRD, SRS, and SDD documents in **PTC Integrity**, a software that records system and software requirements ensuring traceability and compliance (used by the FDA to review new releases).

Software Engineer Intern @ IQM Corporation

Remote | May 2022 - Aug 2022

- Crafted modern, performant, reliable code in **TypeScript**, **React.js**, **Redux**, **SASS** for the firm's flagship project: a suite of ad-campaign management web applications
- Collaborated in an agile scrum environment leveraging **Git**, **BitBucket**, **Jira**, and **Jenkins** for streamlined project management and version control
- Implemented critical UX features and bug fixes, resulting in significant improvements to clients' user experience
- Developed a complete overhaul of the main lead generation website (iqm.com) with **Remix.js** and **Bootstrap**

PROJECTS

(See more at <https://vyas.app/projects>)

PhotoHunter (<https://vyas.app/projects/photohunter>)

- A **React Native** scavenger hunt app similar to Geocaching, where users take photos of specific locations and objects, validated using **multi-modal LLMs**. Maps UI with **Mapbox**, **Django & PostgreSQL** backend, **AWS S3** for photo storage.
- **LLMs**: **Langchain**. Deploying with **AWS Lambda**, **API Gateway**, & **AWS RDS** for database. **CI/CD** with **GitHub Actions**.

ASU Course Tracker (<https://vyas.app/projects/asu-course-tracker>)

- A **React Native** mobile app with a **C# .NET Core REST API** and **PostgreSQL** backend that scrapes ASU data using **Playwright**, and lets students at ASU track the number of seats available for any course and receive notifications.
- Deployed infrastructure with **Docker containers** on **GKE** (Google **Kubernetes Engine**), using **AWS RDS (Postgres)** and **AWS ECR** for image storage. **CI/CD** pipeline built with **GitHub Actions** for automated builds, tests, and deployments.

Playing Bowling with RL (<https://vyas.app/projects/bowling-with-rl>)

- Developed and trained a 7-DOF robotic arm to play bowling in **Mujoco** using the **Stable-Baselines3** implementation of **Proximal Policy Optimization (PPO)**, a **reinforcement learning** algorithm.

A Comprehensive Introduction to Reinforcement Learning (https://vyas.app/reinforcement_learning.pdf)

- Reinforcement learning is an absolutely fascinating subfield of machine learning. So I wrote a tutorial in the style of a research paper providing a comprehensive introduction to its fundamentals.

FRDM KL46Z Robot (<https://youtu.be/f86hxOlkRdE>)

- A robot controlled with NXP's FRDM KL46Z Microprocessor, with software written in **C**. Uses the HC SR04 Ultrasonic sensor to sense objects in front of the robot and avoid obstacles.

EDUCATION

Arizona State University

M.S., Robotics & Autonomous Systems (AI)
B.S., Computer Science

Tempe, AZ

Aug 2024 - May 2025
Aug 2020 - May 2024